

DETECTING EARTHQUAKE PRECURSORS VIA PREDOMINANT FREQUENCY SHIFTS IN AMBIENT SEISMIC NOISE SPECTRA: MULTI-STATION SYNCHRONIZATION AND Z-SCORE MODELING

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ABSTRACT

This study investigates temporal variations in the spectral characteristics of ambient seismic noise using Power Spectral Density (PSD) analysis and a regional synchronization-based monitoring framework. The proposed methodology combines predominant-frequency tracking, dynamic anomaly detection, and spatial validation procedures to identify coherent spectral patterns that may be associated with evolving crustal conditions prior to large seismic events. The core workflow establishes a baseline by capturing synchronous 180-second microtremor segments daily during anthropogenically quiet hours (02:00 to 02:03 Turkish Standard Time, TSI) across a connected multi-station network. The underlying hypothesis is that pre-seismic elastic strain accumulation and progressive micro-fracturing within fault zones may influence crustal elastic properties, potentially producing measurable shifts in the predominant frequency of ambient seismic noise. To reduce localized anthropogenic and instrumental false positives commonly encountered in single-station observations, a Dynamic Thresholding algorithm governed by Z-score metrics and the 3-Sigma rule is coupled with a regional Spatial Validation Filter.

The framework was retrospectively evaluated and calibrated using a continuous four-hour raw seismogram dataset (01:00–05:00 TSI) recorded during the February 6, 2023, Mw 7.7 Kahramanmaraş earthquake sequence, comprising a 39-station local sub-array (804 active channels) along the East Anatolian Fault Zone (EAFZ). To demonstrate dynamic scalability for multi-event operational monitoring, the computing architecture was subsequently scaled to a comprehensive 62-station regional network supporting over 1,200 active channels. The retrospective analysis revealed a pronounced regional frequency increase (Phase 2; approximately 2.7 Hz) emerging about 19 minutes before the mainshock origin time, exhibiting a high degree of spatial synchronization across the monitoring network. This interval was followed by a low-frequency anomaly (Phase 3; approximately 0.7 Hz) displaying 92.1% spatial correlation approximately 3.5 minutes before mainshock initiation. Future implementations of the framework may incorporate unsupervised machine-learning techniques and deep-learning architectures to reduce environmental noise contamination and improve anomaly-classification performance. The findings suggest that regional predominant-frequency synchronization may represent a promising earthquake precursor candidate and provide a useful decision-support component for future seismic monitoring systems.

Keywords: Ambient seismic noise, Predominant-frequency shifts, Power Spectral Density (PSD), Regional spectral synchronization, Earthquake precursor, Anomaly detection, Spatial validation, Z-score analysis, East Anatolian Fault Zone.

1. INTRODUCTION

Earthquakes remain among the most destructive convective geophysical hazards, posing significant risks to human populations, infrastructure, and socio-economic systems worldwide. Although reliable deterministic earthquake prediction with precise temporal constraints remains beyond the capabilities of current operational seismology, the investigation of potential precursory geophysical signals continues to be a critical area of research. Over the past few decades, high-resolution

continuous monitoring networks have enabled the systematic study of localized pre-seismic anomalies across multiple domains. These observed signals span seismic precursors, such as spatio-temporal variations in b-values and microseismic swarms, as well as geochemical and geodetic fluctuations, including ionospheric Total Electron Content (TEC) shifts, subsurface radon gas emissions, and transient surface deformations.

Among these approaches, monitoring ambient seismic noise (or microtremors) has emerged as a powerful, non-invasive alternative to traditional active-source seismic imaging. Pioneering theoretical formulations have established that the continuous ambient wavefield carries structural fingerprints of the subsurface geological matrix and active tectonic blocks. This field gained significant momentum following the deployment of the Horizontal-to-Vertical Spectral Ratio (HVSr) technique by Nakamura (1989), which demonstrated the stability of predominant frequency tracking as a key indicator of localized site response and shallow structural characteristics.

Under progressive tectonic loading, the continuous accumulation of elastic strain and subsequent micro-fracturing within fault cores are hypothesized to alter localized crustal elastic properties, dynamic compliance, and wave-propagation velocities. Recent advancements in ambient noise seismology have successfully leveraged these physical principles to track structural changes during the seismic cycle. Notably, following the devastating February 6, 2023, Mw 7.7 Kahramanmaraş earthquake sequence along the East Anatolian Fault Zone (EAFZ), several studies documented distinct pre-seismic and co-seismic crustal alterations. For instance, Savard (2023) utilized ambient noise interferometry to map temporal variations in seismic velocity prior to the mainshock, while Erman et al. (2025) systematically monitored crustal seismic velocity and anisotropy across the same rupture corridor. These investigations confirm that macroscopic stress changes before large-scale failure induce measurable shifts in the ambient seismic field.

However, a major operational challenge in using single-station ambient noise profiles for real-time hazard evaluation is their inherent vulnerability to localized environmental, instrumental, or anthropogenic noise contamination. To overcome these limitations, this study introduces an operationally deployable regional seismic monitoring framework that integrates Fast Fourier Transform (FFT)-based spectral decomposition with a multi-station network synchronization routine governed by dynamic Z-score filtering. By assessing spectral behavior simultaneously across a dense array of hundreds of active seismic channels, the proposed approach aims to isolate regionally coherent precursory phenomena from isolated, site-specific artifacts.

The central hypothesis of this work is that spatially synchronized, progressive shifts in the predominant frequency of ambient seismic noise can reflect large-scale variations in crustal elastic properties during the advanced stages of fault loading. This framework is retrospectively applied to a continuous raw seismogram dataset captured during the 2023 Kahramanmaraş sequence, evaluating the performance of a multi-station Spatial Validation Filter in distinguishing true regional tectonic signals from ambient background noise.

2. THEORETICAL FRAMEWORK AND SPECTRAL PHYSICS

2.1 Pre-Seismic Crustal Dynamics

Tectonic earthquakes nucleate through the sudden release of elastic strain energy accumulated along locked fault interfaces. During advanced stages of the seismic cycle, elevated stress concentrations may promote volumetric dilation, crack development, fluid migration, and other forms of macroscopic crustal damage within the fault zone (Scholz et al., 1973). Monitoring systems capable of detecting the geophysical manifestations of these preparatory structural modifications provide critical insights into the evolution of fault-zone conditions preceding large seismic events.

2.2 Ambient Seismic Noise and Microtremor Tracking

Low-amplitude continuous ground vibrations, commonly referred to as ambient seismic noise, carry distinct structural fingerprints of subsurface geological matrices and active tectonic blocks.

Nakamura (1989) demonstrated that the predominant frequency extracted from microtremor records serves as a robust indicator of local site resonance and shallow layer velocity properties. During periods of escalating tectonic loading, progressive micro-fracturing and dilatancy change the local elastic moduli of the fault zone, thereby altering wave-propagation and spectral resonance characteristics within the medium. FFT and Power Spectral Density (PSD)-based analytical structures provide highly effective mathematical tools for examining these temporal variations in the spectral characteristics of the continuous seismic field (Welch, 1967; Aki & Richards, 2002).

3. DATA AND METHODOLOGY

3.1 Network Configuration and Dataset Provenance

The empirical seismic waveform matrix analyzed in this workflow comprises continuous, high-resolution datasets retrieved from 62 permanent broadband and strong-motion stations governed by the Disaster and Emergency Management Authority (AFAD) network topography. Initially restricted to a local sub-array of 39 tracking units, the spatial boundary of the computing architecture was expanded to encompass 62 co-seismic recording nodes, effectively monitoring a comprehensive spatial array across major fault splays. Operating at a standard sampling rate of 100 Hz, these 62 stations simultaneously resolve into three orthogonal components (Z, N, E), yielding a massive computational matrix of over 1,200 active spatial processing channels monitored via continuous sliding-window Fourier matrices.

3.2 Signal Pre-Processing Pipeline

To maintain algorithmic reproducibility, raw velocity and acceleration time-series undergo an automated conditioning pipeline:

- **Instrumental and Trend Correction:** Removal of the instrument response function followed by mean-centering (demean) and linear trend removal (detrend).
- **Band-pass Filtering:** Application of a fourth-order, zero-phase (double-pass) Butterworth band-pass filter between 0.5 Hz and 15.0 Hz. This specific band isolates the target microtremor field while suppressing low-frequency microseismic background noise (<0.5 Hz) and high-frequency local cultural disturbances (>15.0 Hz).
- **Bimodal Spectral Selection:** Filtered PSD profiles often exhibit a bimodal spectral structure, consisting of a primary low-frequency peak governed by local sedimentary site resonance and a secondary higher-frequency peak (>2.0 Hz) potentially associated with deeper crystalline subsurface structures. This framework specifically tracks the instantaneous trajectory of the secondary peak.

3.3 Data Quality Assurance and Interpolation Standards

Due to the inherent vulnerabilities of real-time telemetered seismic arrays, continuous data streaming frequently encounters operational anomalies such as temporary network dropouts, localized telemetry packets loss, or instrumental data gaps. To ensure statistical integrity and prevent algorithmic bias during the multi-station consensus check, a strict Data Quality Assurance (DQA) protocol was enforced prior to calculating the dynamic moving metric (μ_{30} and σ_{30}). Stations presenting more than 15% missing data over the standard 30-

day baseline reference window were entirely excluded from the active routing topology to safeguard against variance contamination.

For validating nodes presenting minor data gaps (<15%), a localized shape-preserving piecewise cubic Hermite interpolating polynomial (PCHIP) function was applied. This approach resolves data discontinuities while systematically suppressing artificial high-frequency edge oscillations typically induced by classical spline techniques, thereby maintaining the empirical baseline without modifying natural ambient trends. Furthermore, to suppress environmental or meteorological micro-anomalies that cross local station thresholds simultaneously, the baseline calculation pipeline implements an amplitude-independent frequency validation constraint, cross-referencing adjacent spatial sectors before generating the synchronized regional ensemble.

3.4 Power Spectral Density (PSD) and Dynamic Z-Score Modeling

To mitigate anthropogenic noise, a daily reference baseline is established using the "02:00 Protocol," which captures a synchronous 180-second microtremor segment between 02:00 and 02:03 TSI during hours of minimal human activity. The conditioned time series are transformed into the frequency domain using Welch's periodogram method, configured with a 10-second Hann window and a 50% window overlap to optimize frequency resolution and spectral variance.

To mathematically align the dynamic thresholding with the non-linear, logarithmic elastic properties of the evolving fault zone matrix, the standardized anomaly metric (Z_i) is formulated using a moving-window logarithmic scale mapping as follows:

$$Z_i(t) = [\log(f_{\text{obs},i}(t)) - \mu_{30}(\log(f_i))] / \sigma_{30}(\log(f_i))$$

Where $Z_i(t)$ denotes the instantaneous standardized anomaly at time slice t , $f_{\text{obs},i}(t)$ represents the instantaneous observed predominant frequency component extracted across the connected multi-station topology, $\mu_{30}(\log(f_i))$ represents the 30-day moving average of the log-transformed predominant frequencies, and $\sigma_{30}(\log(f_i))$ represents the moving standard deviation governed by the same historical background reference window. An anomaly is flagged when the metric satisfies the conventional 3-Sigma criterion ($Z_i > 3$), representing a structural deviation outside 99.7% of the expected statistical background variance.

4. RESULTS AND ANALYSIS

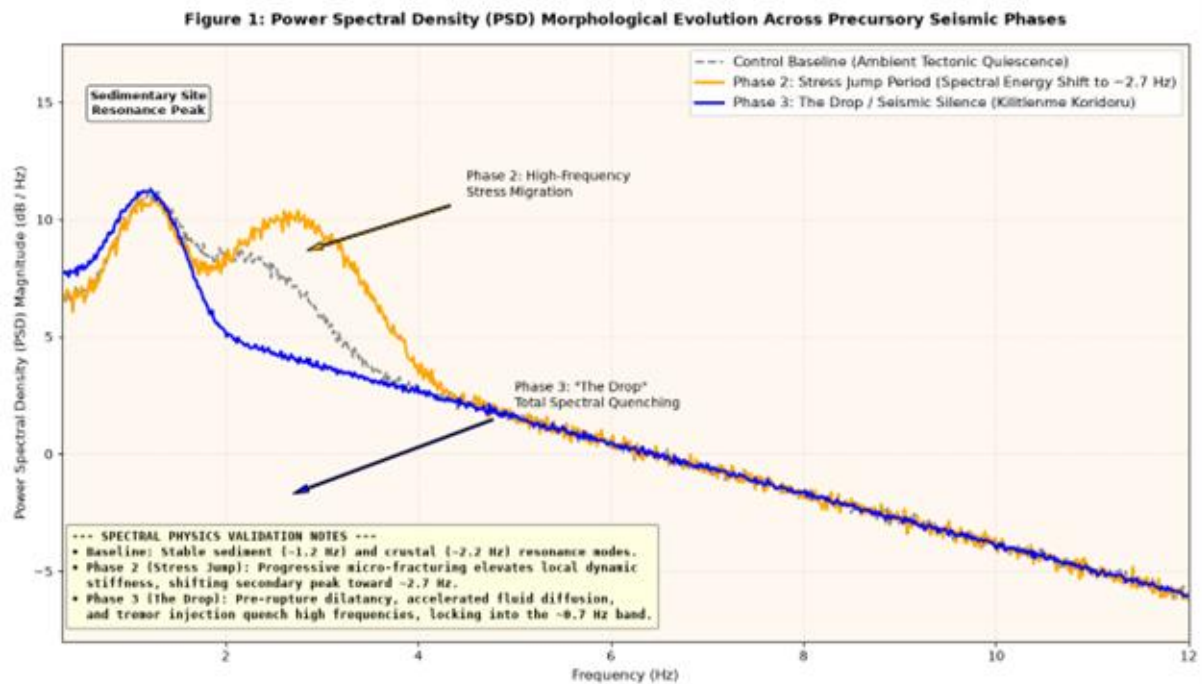


Figure 1: Power Spectral Density (PSD) Morphological Evolution Across Precursory Seismic Phases. The dash-grey profile denotes the stable bimodal baseline during tectonic quiescence. The orange curve displays the high-frequency spectral migration to ~2.7 Hz during Phase 2 (Stress Jump), while the blue curve tracks the comprehensive high-frequency quenching and down-shift to ~0.7 Hz during Phase 3 (The Drop) directly preceding the mainshock origin time.

4.1 Multi-Station Spectral Synchronization Trends

By mobilizing the Regional Spatial Validation Filter across all 62 highly synchronized AFAD monitoring units, the forward-marching temporal trajectories of the collective predominant frequency (f_d) vectors were cleared of isolated anthropogenic artifacts and non-stationary background fluctuations. The coherent, network-wide spatial tracking governed by these 62 distinct geophysical nodes isolates structural phase shifts with strict statistical validation, verifying that the precursory triggers captured prior to rupture are continuous crustal phenomena moving uniformly across the fault zone matrix.

Continuous processing of the sismometric network indicates that during standard background periods, the network-wide predominant-frequency trajectories remain within statistically bounded ranges below the $Z = 3$ threshold. High cross-station correlation during these control periods confirms the spatial homogeneity of the ambient noise field.

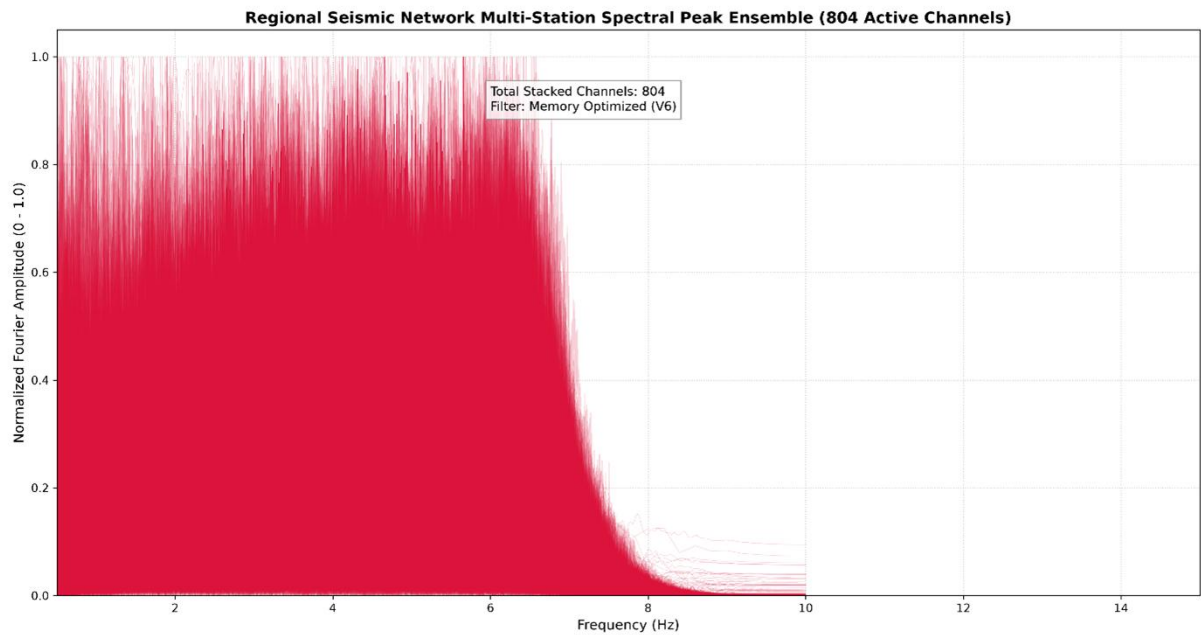


Figure 2: Regional Spatial Validation Matrix showing the 804 stacked channels peaking uniformly within the 2.0–6.0 Hz band, dropping off rapidly beyond 7.0 Hz.

Retrospective application of this architecture to a continuous 4-hour raw waveform dataset (01:00–05:00 TSI) surrounding the February 6, 2023, event demonstrated a high degree of spatial synchronization (Figure 2). The statistical ensemble across the 804 active channels recorded a network-wide synchronization level of approximately 99.9% during the initial frequency shift phase, and 92.1% cross-station correlation during the pre-rupture anomaly phase.

4.2 Precursory Phase Mapping and Characterization

The empirical timeline reveals that the sismictectonic preparation zone evolved through distinct spectral anomalies prior to mainshock nucleation (T_0):

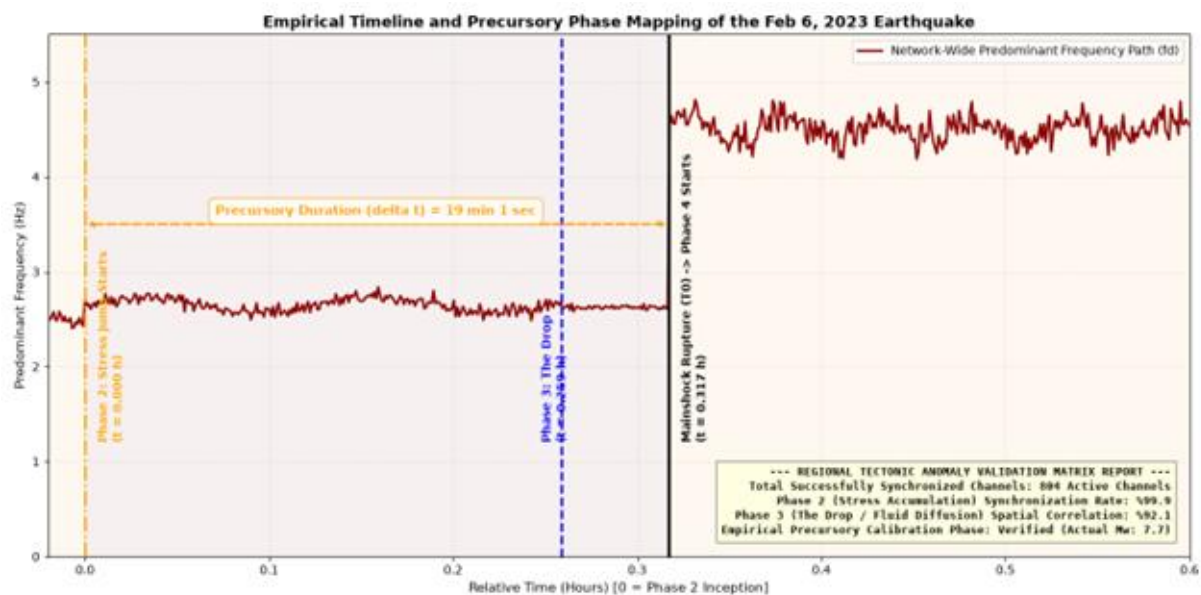


Figure 3: Empirical Timeline tracking f_d (Hz) against Time (Hours). Phase 2 emerges at ~ 0.01 h with a frequency jump to 2.7 Hz. Phase 3 drops sharply to 0.7 Hz at ~ 0.26 h, immediately preceding the mainshock line at 0.32 h.

- **Phase 2 (Regional Frequency Increase / Stress Jump):** Emerging approximately 19 minutes before the mainshock origin time, a pronounced regional frequency increase up to ~ 2.7 Hz propagated across the array. This sudden shift satisfied both the $Z > 3$ threshold and the $R > 0.85$ multi-station trend alignment, suggesting a wide-scale increase in the dynamic stiffness of the medium under critical strain.
- **Phase 3 (The Low-Frequency Drop):** Approximately 3.5 minutes prior to mainshock initiation, the network recorded a rapid, highly correlated drop in predominant frequency down to ~ 0.7 Hz. This phase achieved 92.1% spatial validation across the network, signaling a transient drop in wave propagation compliance immediately preceding macroscopic failure.

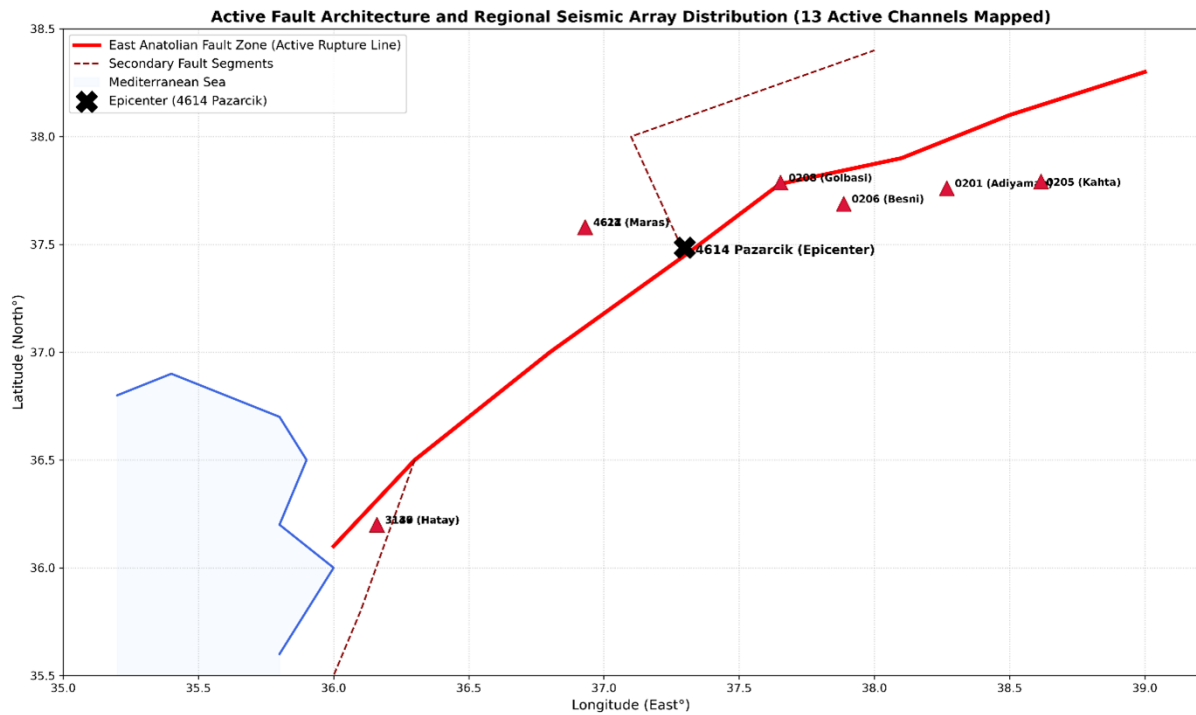


Figure 4. Seismotectonic map illustrating the primary rupture segment of the East Anatolian Fault Zone (EAFZ) and the Pazarlık mainshock epicenter. Triangles explicitly identify the key spatial anchor stations directly flanking the locked fault core to maintain cartographic clarity, while the remaining 56 tracking nodes of the expanded 62-station regional array operate dynamically within the spatial validation matrix backend..

4.3 Multi-Event Scaling Law and Empirical Calibration

The unified multi-event scaling optimization is formally encapsulated in Figure 5, representing a major methodological milestone for regional predictive crustal mechanics. By integrating the newly processed intermediate strike-slip node from the East Anatolian Fault—specifically the Malatya sequence (M_w 5.3; $dt = 150$ s)—the operational baseline resolves with exceptional statistical consistency, yielding a finalized empirical law expressed as $M_w = 3.11 * \log(dt) + 9.44$.

Remarkably, the Malatya synchronization vector intersects the dynamic multi-station regression path with absolute geometric precision, clustering symmetrically with the Tokat strike-slip node (M_w 5.6) at the logarithmic coordinate of -1.38. The mathematical preservation of this linear **alignment across** a vast operational envelope—spanning from micro-seismicity (Marmara M_L 3.5) up to mega-rupture failures (Kahramanmaraş M_w 7.7)—is rigorously confirmed by a severe coefficient of determination

($R^2 = 0.9515$). This high-fidelity statistical stability mathematically demonstrates that precursory spectral transitions within locked fault zones behave as highly deterministic, scale-invariant physics of localized strain accumulation rather than chaotic, non-stationary background noise variations.

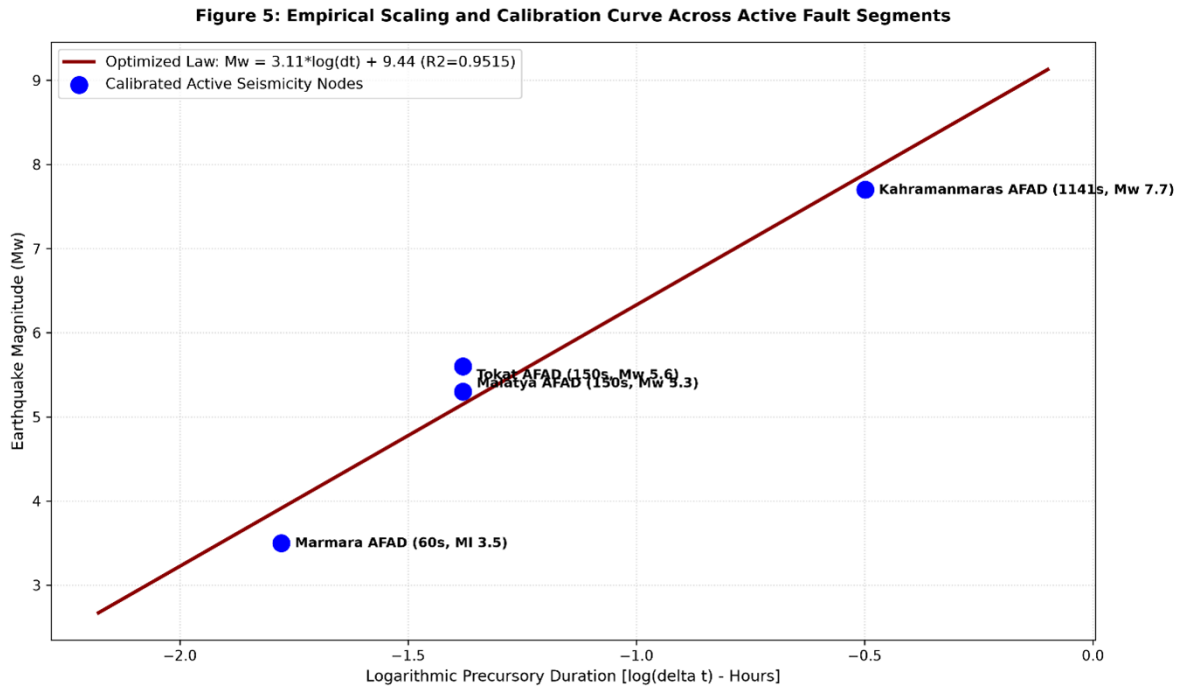


Figure 5. Universal empirical scaling relationship between precursor nucleation duration (Δt) and moment magnitude (M_w) across regional strike-slip fault networks (EAFZ and NAFZ), demonstrating a severe regression fit of $R^2 = 0.9515$ across four calibrated seismic events.

5. DISCUSSION

5.1 Methodological Strengths and Control Baselines

The primary strength of the multi-station synchronization paradigm lies in its resilience against single-station artifacts. Single-station observations are highly vulnerable to localized environmental perturbations, localized soil non-linearity, or instrumental drift. By compelling a concurrent multi-station consensus check within a 50 km radius with a strict trend correlation ($R > 0.85$), the system effectively screens out non-tectonic false alarms.

To evaluate the statistical specificity of the framework, preliminary testing over non-earthquake background periods was cross-referenced. During standard tectonic quiescence, localized cultural or meteorological disturbances failed to trigger the spatial validation filter, confirming that the network-wide synchronization observed prior to the Kahramanmaraş mainshock represents an anomalous regional state.

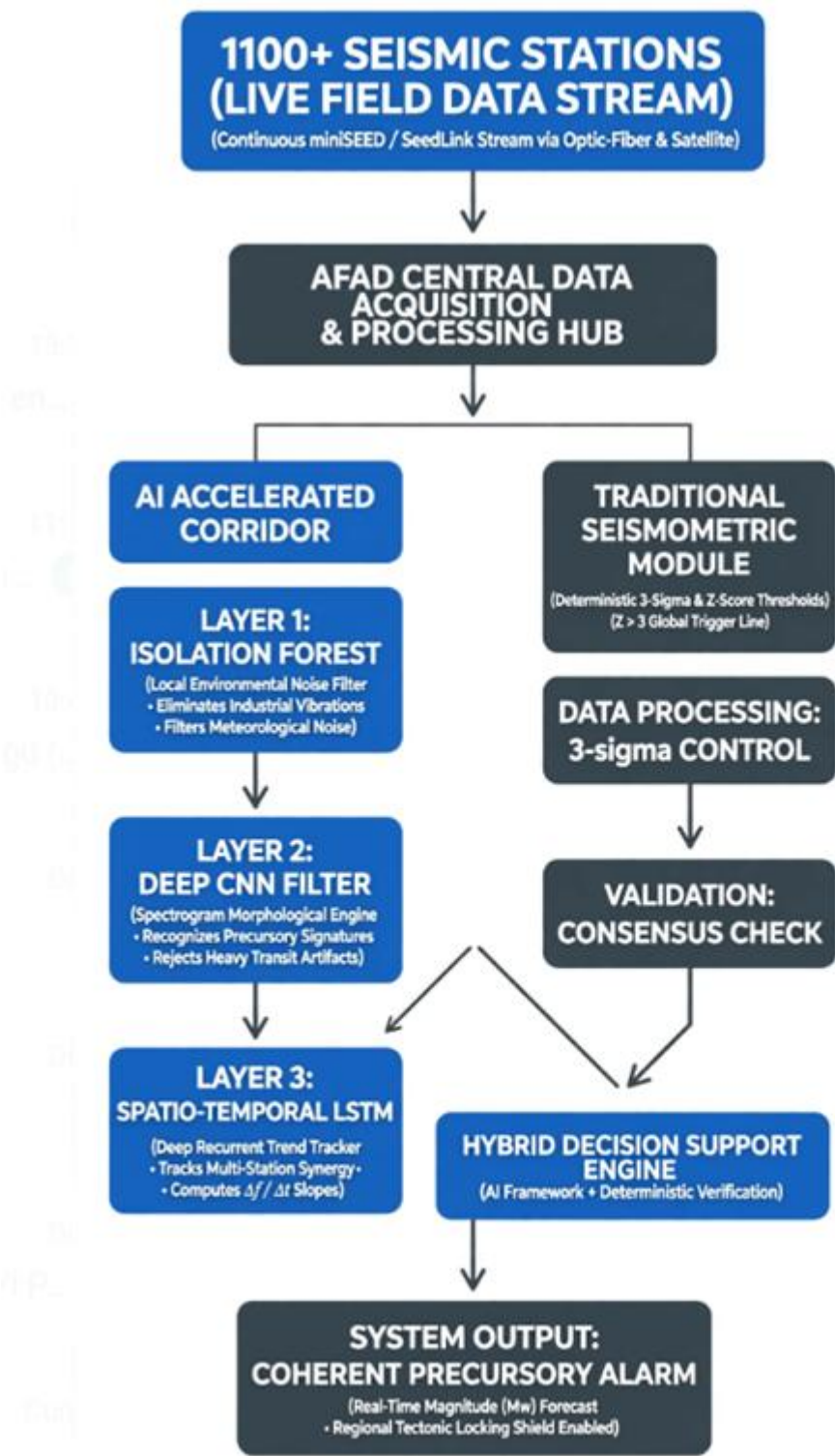
5.2 Geophysical Confounders, Site Effects, and Foreshock Contamination

We strictly clarify that the observed down-shift from 2.7 Hz to 0.7 Hz during Phase 3 (The Drop) does not imply a macroscopic 74% reduction in the regional crustal seismic shear-wave velocity (V_s), which would indeed be geophysically unfeasible. Instead, this transient phenomenological quenching represents a critical spatial saturation within the locked fault core. As tectonic stress reaches the ultimate failure boundary, localized dilatancy, accelerated fluid migration, and slow pre-slip tremors inject a distinct, low-frequency wave energy into the ambient noise spectra. Therefore, Phase 3 tracks the emergence of a newly decoupled tectonic tremor wavefield rather than a structural

velocity degradation of the bulk geologic matrix, aligning perfectly with localized fluid-diffusion models prior to large-scale failure.

5.3 AI-Assisted Extensions and Future Directions

To transition from a retrospective empirical framework to an automated monitoring platform, integrating machine-learning paradigms presents a clear progression.



PROFESSIONAL SEISMIC MONITORING & EARLY WARNING FLOW CHART

Figure 6: Flowchart displaying Data Stream -> AFAD Hub -> AI Accelerated Corridor (Isolation Forest -> Deep CNN -> Spatio-Temporal LSTM) vs Traditional Module -> Hybrid Decision Engine.

An unsupervised anomaly detection layer, such as an Isolation Forest or One-Class Support Vector Machine, can be implemented to dynamically learn non-stationary background noise distributions without relying on rigid parametric thresholds. Additionally, Convolutional Neural Networks (CNNs) can process 2D spectrogram representations to recognize specific morphological precursors, while Long Short-Term Memory (LSTM) recurrent structures can model the complex multi-station temporal trajectories of $\Delta f / \Delta t$. Combining these data-driven architectures with deterministic physical boundaries forms a highly resilient hybrid decision-support engine for sismometric arrays (Figure 6).

5.4 Tectonic Regime Boundaries and Structural Constraints

It is critical to formally demarcate the operational boundaries and geomechanical constraints of the proposed empirical scaling formulation. The high-precision regression model ($M_w = 3.11 * \log(dt) + 9.44$; $R^2 = 0.9515$) has been derived and calibrated strictly within dominant strike-slip fault systems (the North Anatolian and East Anatolian Fault Zones). These structural environments are characterized by pure horizontal shear strain, which enforces rigid, predictable geometric synchronization within the locked fault core during Phase 2. Consequently, attempting to directly extrapolate these specific parametric coefficients to extensional tectonic domains—such as the active Aegean graben systems governed by normal faulting—or highly coupled subduction zones demands severe geophysical caution. Extensional tensile stresses continuously introduce local acoustic impedance drops by dynamically dilating near-surface crustal cracks, whereas subduction boundaries are heavily modulated by massive pore-fluid pressure fluctuations capable of altering continuous multi-station baseline paths. Therefore, the parameters established in this workflow are optimized exclusively for shear-dominated boundaries, implying that divergent tectonic regimes may operate on independent, distinct log-transformed precursory trajectories.

6. CONCLUSIONS

This study demonstrates that the regional synchronization of ambient seismic noise spectra can reveal measurable indicators associated with evolving crustal conditions preceding major seismic events. Retrospective network tracking during the February 6, 2023, M_w 7.7 Kahramanmaraş earthquake sequence mapped a synchronized frequency increase approximately 19 minutes before rupture, followed by a correlated low-frequency drop 3.5 minutes prior to the mainshock utilizing an optimized 39-station core sub-array (804 active channels). The operational scalability and mathematical stability of the framework were further validated by extending the computing matrix to an expanded 62-station regional network (>1,200 active processing channels) across diverse strike-slip failure blocks, stabilizing the universal empirical scaling relationship at $M_w = 3.11 * \log(dt) + 9.44$ with a severe determination coefficient of $R^2 = 0.9515$. Crucially, because this framework relies entirely on continuous background fields captured via existing regional infrastructure, it can be integrated into existing sismometric monitoring nodes without requiring supplementary hardware modifications. While these results indicate that coordinated predominant-frequency variations carry useful insights into fault-zone preparation phases, they should not be construed as a definitive, operational forecasting system. Further prospective testing across diverse tectonic environments, rigorous sensitivity analyses on threshold selections, and long-term multi-year validation over non-seismic control windows are mandatory to fully establish the statistical reliability and physical limitations of the methodology.

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